DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Subject Name: **ARTIFICIAL INTELLIGENCE** Subject Code: **CS T71**

UNIT III

Reasoning under uncertainty: Logics of non-monotonic reasoning-Implementation- Basic probability notation - Bayes rule – Certainty factors and rule based systems-Bayesian networks – Dempster - Shafer Theory - Fuzzy Logic.

11 Marks

1. What is uncertainty? Explain

Uncertainty

Let action At = leave for airport t minutes before flight

Will At get me there on time?

Problems:

1) partial observability (road state, other drivers' plans, etc.)

2) noisy sensors (KCBS tra_c reports)
3) uncertainty in action outcomes (at tire, etc.)

4) immense complexity of modelling and predicting tra_c Hence a purely logical approach either

1) risks falsehood: \A25 will get me there on time"

or 2) leads to conclusions that are too weak for decision making:
"A25 will get me there on time if there's no accident on the bridge and it doesn't rain and my tires remain intact etc etc."

Methods for handling uncertainty

Default or nonmonotonic logic:

Assume my car does not have a at tire

Assume A25 works unless contradicted by evidence

Issues: What assumptions are reasonable? How to handle contradiction?

Rules with fudge factors:

$$A_{25} \mapsto_{0.3} AtAirportOnTi$$
$$Sprinkler \mapsto_{0.99} WetGra$$

Issues: Problems with combination, e.g., Sprinkler causes Rain??

1. What is non-monotonic reasoning? Explain? (Apr-May '14)(Nov – Dec '14)(Nov'15)(Apr-May'17)

A **non-monotonic logic** is a formal logic whose consequence relation is not monotonic. Most studied formal logics have a monotonic consequence relation, meaning that adding a formula to a theory never produces a reduction of its set of consequences. Intuitively, monotonicity indicates that learning a new piece of knowledge cannot reduce the set of what is known.

A monotonic logic cannot handle various reasoning tasks such as reasoning by default (consequences may be derived only because of lack of evidence of the contrary), adductive reasoning (consequences are only deduced as most likely explanations), some important approaches to reasoning about knowledge

Default reasoning

An example of a default assumption is that the typical bird flies. As a result, if a given animal is known to be a bird, and nothing else is known, it can be assumed to be able to fly. The default assumption must however be retracted if it is later learned that the considered animal is a penguin. This example shows that a logic that models default reasoning should not be monotonic.

Logics formalizing default reasoning can be roughly divided in two categories: logics able to deal with arbitrary default assumptions (default logic, defeasible logic/defeasible reasoning/argument (logic), and answer set programming) and logics that formalize the specific default assumption that facts that are not known to be true can be assumed false by default (closed world assumption and circumscription).

Abductive reasoning

Abductive reasoning is the process of deriving the most likely explanations of the known facts. An abductive logic should not be monotonic because the most likely explanations are not necessarily correct.

For example, the most likely explanation for seeing wet grass is that it rained; however, this explanation has to be retracted when learning that the real cause of the grass being wet was a sprinkler. Since the old explanation (it rained) is retracted because of the addition of a piece of knowledge (a sprinkler was active), any logic that models explanations is non-monotonic.

Reasoning about knowledge

If a logic includes formulae that mean that something is not known, this logic should not be monotonic. Indeed, learning something that was previously not known leads to the removal of the formula specifying that this piece of knowledge is not known. This second change (a removal caused by an addition) violates the condition of monotonicity. A logic for reasoning about knowledge is the auto epistemic logic.

Belief revision

Belief revision is the process of changing beliefs to accommodate a new belief that might be inconsistent with the old ones. In the assumption that the new belief is correct, some of the old ones have to be retracted in order to maintain consistency. This retraction in response to an addition of a new belief makes any logic for belief revision to be non-monotonic. The belief revision approach is alternative to para consistent logics, which tolerate inconsistency rather than attempting to remove it.

3. What is probability and Basic probability notation? (Apr-May '15)_P

robability

Given the available evidence, A25 will get me there on time with probability 0:04

(Fuzzy logic handles degree of truth NOT uncertainty e.g., WetGrass is true to degree 0:2)

Probabilistic assertions summarize effects of

laziness: failure to enumerate exceptions, qualifications, etc.

ignorance: lack of relevant facts, initial conditions, etc.

Subjective or Bayesian probability:

Probabilities relate propositions to one's own state of knowledge e.g., P(A25/no reported accidents) = 0:06

These are not claims of a \probabilistic tendency" in the current situation (but might be learned from past experience of similar situations)

Probabilities of propositions change with new evidence:

e.g., P(A25jno reported accidents; 5 a.m.) = 0:15

(Analogous to logical entailment status KB j= _, not truth.)

Making decisions under uncertainty

Suppose I believe the following:

P(A25 gets me there on timej : : :) = 0:04

P(A90 gets me there on timej : : :) = 0:70

P(A120 gets me there on timej : : :) = 0:95

P(A1440 gets me there on timej : : :) = 0:9999

Which action to choose?

Depends on my preferences for missing ight vs. airport cuisine, etc.

Utility theory is used to represent and infer preferences

Decision theory = utility theory + probability theory

Probabilistic Reasoning

Using logic to represent and reason we can represent knowledge about the world with facts and rules, like the following ones:

bird(tweety).

fly(X) :- bird(X).

We can also use a theorem-prover to reason about the world and deduct new facts about the world, for e.g.,

?- fly(tweety).

Yes

However, this often does not work outside of toy domains - non-tautologous certain rules are hard to find. A way to handle knowledge representation in real problems is to extend logic by using certainty factors. In other words, replace

IF condition THEN fact with

IF condition with certainty *x* THEN fact with certainty *f(x)* Unfortunately cannot really adapt logical inference to probabilistic inference, since the latter is not context-free. Replacing rules with conditional probabilities makes inferencing simpler.

Replace smoking -> lung cancer

or

lots of conditions, smoking -> lung cancer

with

P(lung cancer | smoking) = 0.6

Uncertainty is represented explicitly and quantitatively within probability theory, a formalism that has been developed over centuries.

A probabilistic model describes the world in terms of a set S of possible states - the sample space. We don't know the true state of the world, so we (somehow) come up with a probability distribution over S which gives the probability of any state being the true one.

The world usually described by a set of variables or attributes. Consider the probabilistic model of a fictitious medical expert system. The 'world' is described by 8 binary valued variables:

Visit to Asia? A

Tuberculosis? T

Either tub. or

lung cancer? E

Lung cancer? L

Smoking? S

Bronchitis? B

Dyspnoea? D

Positive X-ray? X

We have 2^8 = 256 possible states or configurations and so 256 probabilities to find. 10.3 Review of Probability Theory .The primitives in probabilistic reasoning are *random variables*. Just like primitives in Propositional Logic are propositions.

A random variable is not in fact a variable, but a function from a sample space S to another space, often the real numbers. For example, let the random variable Sum (representing outcome of two die throws) be defined thus:

$$Sum(\text{die1}, \text{die2}) = \text{die1} + \text{die2}$$

Each random variable has an associated probability distribution determined by the underlying distribution on the sample space

Continuing our example : P(Sum = 2)

= 1/36, P(Sum = 3) = 2/36, . . . ,

P(Sum = 12) = 1/36

Consider the probabilistic model of the fictitious medical expert system mentioned before. The sample space is described by 8 binary valued variables.

Visit to Asia? A

Tuberculosis? T

Either tub. or

lung cancer? E

Lung cancer? L

Smoking? S

Bronchitis? B

Dyspnoea? D

Positive X-ray?

X

There are 2^8 = 256 events in the sample space. Each event is determined by a joint instantiation of all of the variables.

S = {(A = f, T = f,E = f,L = f, S = f,B = f,D = f,X = f), (A = f, T = f,E = f,L = f, S = f,B = f,D = f,X = t), . . .

(A = t, T = t,E = t,L = t, S = t,B = t,D = t,X = t)}

Since S is defined in terms of joint instantations, any distribution defined on it is called a joint distribution. ll underlying distributions will be joint distributions in this module. The variables {A,T,E, L,S,B,D,X} are in fact random variables, which ‘project’ values.

L(A = f, T = f,E = f,L = f, S = f,B = f,D = f,X = f) = f

L(A = f, T = f,E = f,L = f, S = f,B = f,D = f,X = t) = f

L(A = t, T = t,E = t,L = t, S = t,B = t,D = t,X = t) = t

Each of the random variables {A, T, E, L, S, B, D, X} has its own distribution, determined by the underlying joint distribution. This is known as the margin distribution. For example, the distribution for L is denoted P(L), and this distribution is defined by the two probabilities P(L = f) and P(L = t). For example,

P(L = f)

= P(A = f, T = f,E = f,L = f, S = f,B = f,D = f,X = f)

+ P(A = f, T = f,E = f,L = f, S = f,B = f,D = f,X = t)

+ P(A = f, T = f,E = f,L = f, S = f,B = f,D = t,X = f)

. . .

P(A = t, T = t,E = t,L = f, S = t,B = t,D = t,X = t) P(L) is an example of a marginal distribution.

5. Explain Bayes' Rule

Product rule $P(a \wedge b) = P(a|b)P(b) = P(b|a)P(a)$

$\Rightarrow$ Bayes' rule $P(a|b) = \dfrac{P(b|a)P(a)}{P(b)}$

or in distribution form

$$\mathbf{P}(Y|X) = \frac{\mathbf{P}(X|Y)\mathbf{P}(Y)}{\mathbf{P}(X)} = \alpha\mathbf{P}(X|Y)\mathbf{P}(Y)$$

Useful for assessing diagnostic probability from causal

$P(Cause|Effect) = \frac{P(Effect|Cause)P(Cau}{}$

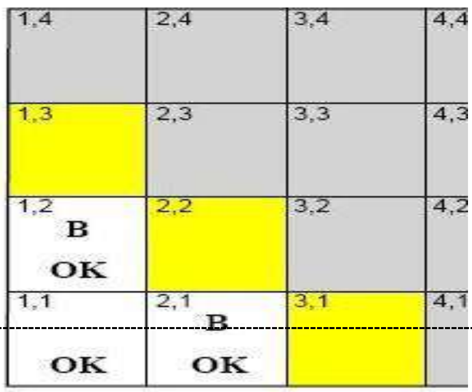
$$\mathbf{P}(Cavity|toothache \wedge catch)$$
$$= \alpha\, \mathbf{P}(toothache \wedge catch|Cavity)\mathbf{P}(Cavity)$$
$$= \alpha\, \mathbf{P}(toothache|Cavity)\mathbf{P}(catch|Cavity)\mathbf{P}(Cavity)$$

This is an example of a naive Bayes model:

$$\mathbf{P}(Cause, Effect_1, \ldots, Effect_n) = \mathbf{P}(Cause)\Pi_i \mathbf{P}(Ef$$

Bayes' Rule and conditional independence

Wumpus World



Specifying the probability model

The full joint distribution is $\mathbf{P}(P_{1,1}, \ldots, P_{4,4}, B_{1,1}, B_{1,2}, B$

Apply product rule: $\mathbf{P}(B_{1,1}, B_{1,2}, B_{2,1} \mid P_{1,1}, \ldots, P_{4,4})\mathbf{P}(F$

(Do it this way to get $P(Effect|Cause)$.)

First term: 1 if pits are adjacent to breezes, 0 otherwise

Second term: pits are placed randomly, probability 0.2 per

$$\mathbf{P}(P_{1,1}, \ldots, P_{4,4}) = \prod_{i,j=1,1}^{4,4} \mathbf{P}(P_{i,j}) = 0.2^n \times 0.8^{16-n}$$

Observations and query

We know the following facts:

$b = \neg b_{1,1} \wedge b_{1,2} \wedge b_{2,1}$

$known = \neg p_{1,1} \wedge \neg p_{1,2} \wedge \neg p_{2,1}$

Query is $\mathbf{P}(P_{1,3}|known, b)$

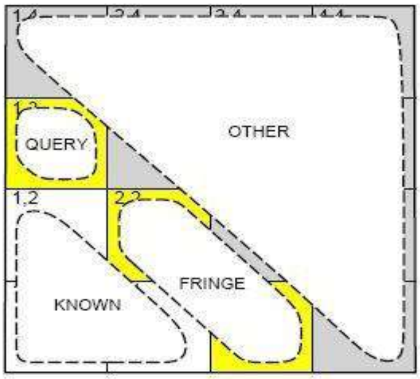
Define $Unknown = P_{ij}$s other than $P_{1,3}$ and $Known$

For inference by enumeration, we have

$$\mathbf{P}(P_{1,3}|known, b) = \alpha \Sigma_{unknown} \mathbf{P}(P_{1,3}, unknown, kn$$

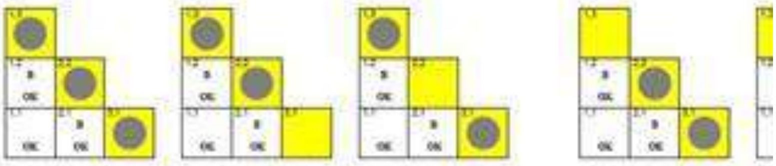
Using conditional independence

Basic insight: observations are conditionally independent of other hidden squares given neighboring hidden squares



Manipulate query into a form where we can use this!

$$\mathbf{P}(P_{1,3}|known, b) = \alpha \sum_{unknown} \mathbf{P}(P_{1,3}, unknown, known, b)$$
$$= \alpha \sum_{unknown} \mathbf{P}(b|P_{1,3}, known, unknown)\mathbf{P}(P_{1,3}, known, unkno$$
$$= \alpha \sum_{fringe} \sum_{other} \mathbf{P}(b|known, P_{1,3}, fringe, other)\mathbf{P}(P_{1,3}, known,$$
$$= \alpha \sum_{fringe} \sum_{other} \mathbf{P}(b|known, P_{1,3}, fringe)\mathbf{P}(P_{1,3}, known, fringe$$
$$= \alpha \sum_{fringe} \mathbf{P}(b|known, P_{1,3}, fringe) \sum_{other} \mathbf{P}(P_{1,3}, known, fringe,$$
$$= \alpha \sum_{fringe} \mathbf{P}(b|known, P_{1,3}, fringe) \sum_{other} \mathbf{P}(P_{1,3})P(known)P(f$$
$$= \alpha\, P(known)\mathbf{P}(P_{1,3}) \sum \mathbf{P}(b|known, P_{1,3}, fringe)P(fring$$



$$\mathbf{P}(P_{1,3}|known, b) = \alpha' \langle 0.2(0.04 + 0.16 + 0.16),\ 0.8(0.04 +$$
$$\approx \langle 0.31, 0.69 \rangle$$

6. Explain Bayesian networks (Nov'13)(Nov'15)

A simple, graphical notation for conditional independence assertions and hence for compact specification of full joint distributions

Syntax:

a set of nodes, one per variable

a directed, acyclic graph (link _ \directly inuences")

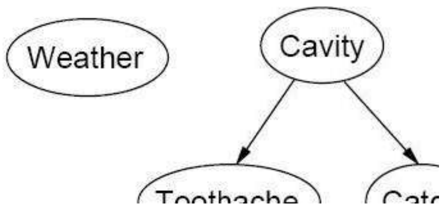
a conditional distribution for each node given its parents:

$$\mathbf{P}(X_i|Parents(X$$

In the simplest case, conditional distribution represented as a conditional probability table (CPT) giving the distribution over Xi for each combination of parent values

Example

Topology of network encodes conditional independence assertions:



Weather is independent of the other variables

Toothache and Catch are conditionally independent given Cavity

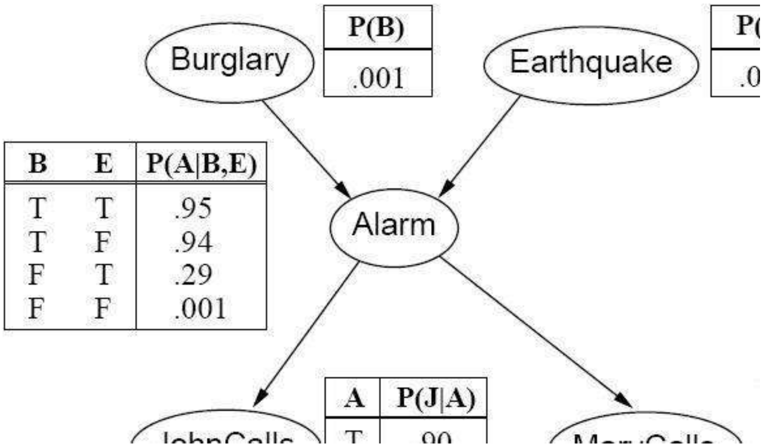
Example

I'm at work, neighbor John calls to say my alarm is ringing, but neighbor Mary doesn't call. Sometimes it's set of by minor earthquakes. Is there a burglar?

Variables: Burglar, Earthquake, Alarm, JohnCalls, MaryCalls

Network topology reects \causal" knowledge:

- A burglar can set the alarm off
- An earthquake can set the alarm off
- The alarm can cause Mary to call
- The alarm can cause John to call



Compactness

A CPT for Boolean X_i with k Boolean parents has 2^k rows for the combinations of parent values

Each row requires one number p for $X_i = true$ (the number for $X_i = false$ is just $1 - p$)

If each variable has no more than k parents, the complete network requires $O(n \cdot 2^k)$ numbers

I.e., grows linearly with n, vs. $O(2^n)$ for the full joint distributic

Constructing Bayesian networks

Need a method such that a series of locally testable assertions of conditional independence guarantees the required global semantics

1. Choose an ordering of variables $X_1, \ldots, X_n$
2. For $i = 1$ to n
   add X_i to the network
   select parents from $X_1, \ldots, X_{i-1}$ such that
   $\mathbf{P}(X_i|Parents(X_i)) = \mathbf{P}(X_i|X_1, \ldots, X_{i-1})$

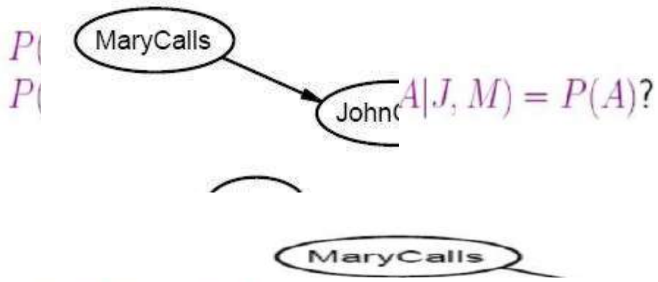
This choice of parents guarantees the global semantics:

$$\mathbf{P}(X_1, \ldots, X_n) = \prod_{i=1}^{n} \mathbf{P}(X_i|X_1, \ldots, X_{i-1}) \quad \text{(chain ru}$$
$$= \prod_{i=1}^{n} \mathbf{P}(X_i|Parents(X_i)) \quad \text{(by constr}$$

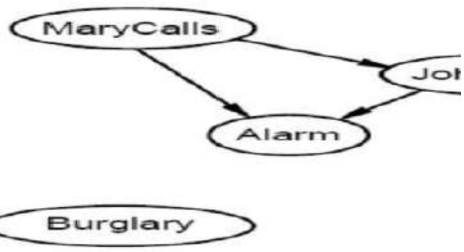
Example

Suppose we choose the ordering M, J, A,

$P(J|M) = P(J$

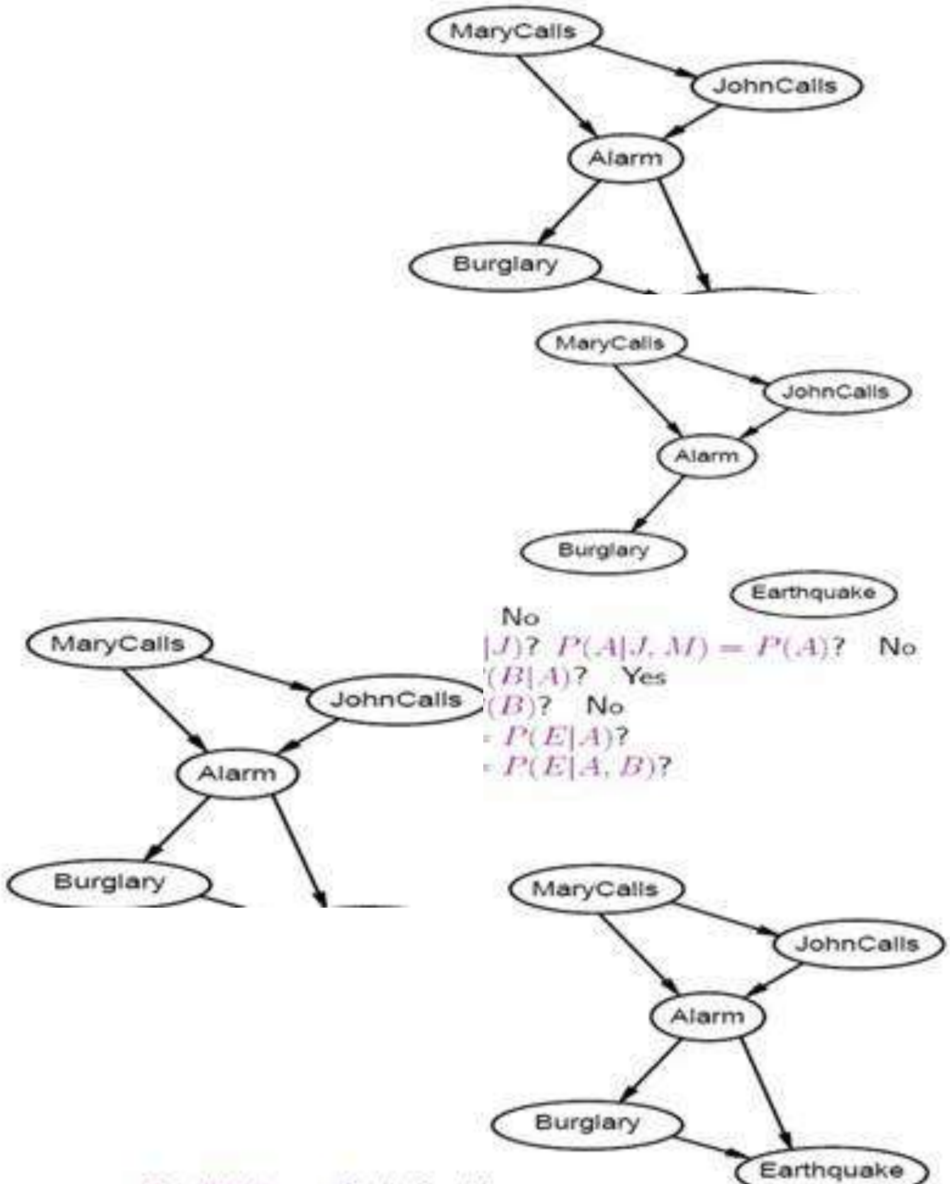


$P(J|M) = P(J)$? No
$P(A|J, M) = P(A|J)$? $P(A|J, M) = P(A)$?



$P(J|M) = P(J)$? No

$P(B|A,J,M) = P(B)$?   No
$P(E|B,A,J,M) = P(E|A)$?
$P(E|B,A,J,M) = P(E|A,B)$?



$P(J|M) = P(J)$?   No

Deciding conditional independence is hard in noncausal directions (Causal models and conditional independence seem hardwired for humans!)

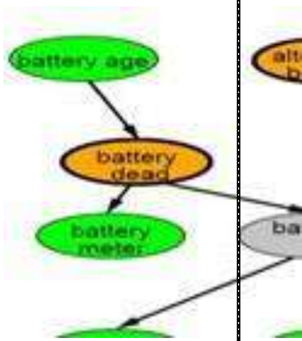
Assessing conditional probabilities is hard in noncausal directions Network is less compact: 1 + 2 + 4 + 2 + 4=13 numbers needed

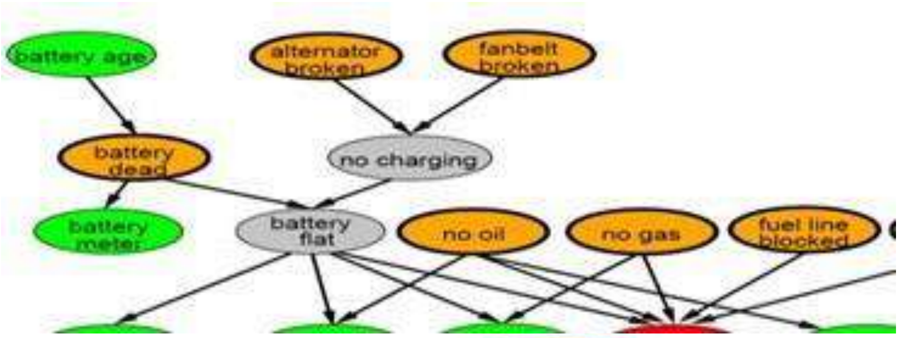
Example: Car diagnosis

Initial evidence: car won't start

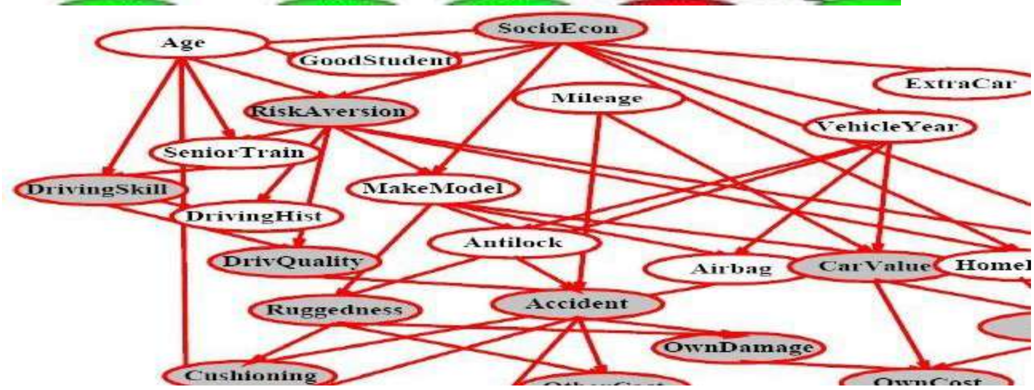
Testable variables (green), \broken, so _x it" variables (orange)

Hidden variables (gray) ensure sparse structure, reduce parameters





Example: Car insurance



Compact conditional distributions

CPT grows exponentially with number of parents
CPT becomes infinite with continuous-valued parent or

Solution: canonical distributions that are defined compa

Deterministic nodes are the simplest case:
$X = f(Parents(X))$ for some function f

E.g., Boolean functions
$NorthAmerican \Leftrightarrow Canadian \vee US \vee Mexic$

E.g., numerical relationships among continuous variable

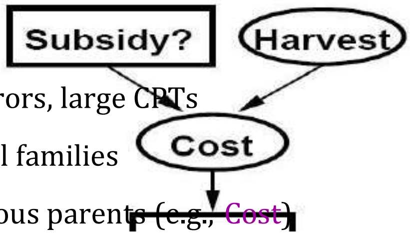
$$\frac{\partial Level}{\partial t} = \text{inflow + precipitation - outflow - evapor}$$

Noisy-OR distributions model multiple noninteracting causes
1) Parents $U_1 \ldots U_k$ include all causes (can add leak node)
2) Independent failure probability q_i for each cause alone
$\Rightarrow P(X|U_1 \ldots U_j, \neg U_{j+1} \ldots \neg U_k) = 1 - \prod_{i=1}^{j} q_i$

$Cold$	Flu	$Malaria$	$P(Fever)$	$P(\neg Fever)$
F	F	F	**0.0**	1.0
F	F	T	0.9	**0.1**
F	T	F	0.8	**0.2**

Hybrid (discrete+continuous) networks

Discrete ($Subsidy?$ and $Buys?$); continuous ($Harvest$ a



Option 1: discretization|possibly large errors, large CPTs

Option 2: finitely parameterized canonical families

1) Continuous variable, discrete+continuous parents (e.g., Cost)
2) Discrete variable, continuous parents (e.g., Buys?)

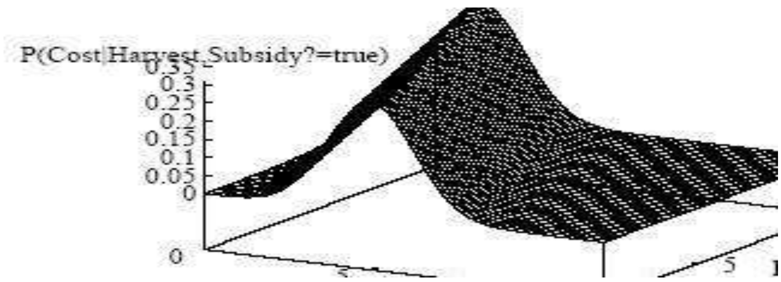
Continuous child variables

Need one conditional density function for child variable given continuous parents, for each possible assignment to discrete parents

Most common is the linear Gaussian model, e.g.,:

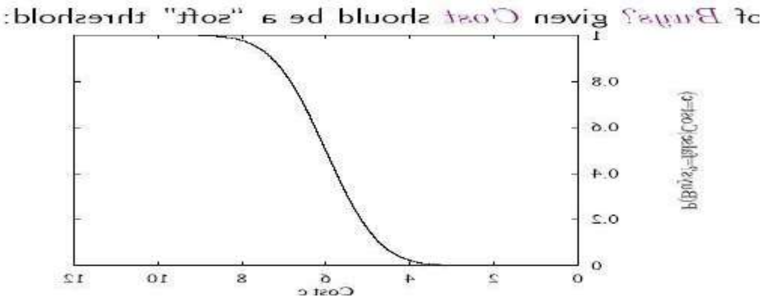
$$
\begin{aligned}
&P(Cost = c | Harvest = h, Subsidy? = \\
&= N(a_t h + b_t, \sigma_t)(c) \\
&= \frac{1}{\sigma_t \sqrt{2\pi}} exp\left(-\frac{1}{2}\left(\frac{c-(a_t h + b_t)}{\sigma_t}\right)^2\right.
\end{aligned}
$$

Mean Cost varies linearly with Harvest, variance is fixed Linear variation is unreasonable over the full range but works OK if the likely range of Harvest is narrow



All-continuous network with LG distributions⇒ full joint distribution is a multivariate Gaussian Discrete+continuous LG network is a conditional Gaussian network i.e., a multivariate Gaussian over all continuous variables for each combination of discrete variable values

Discrete variable w/ continuous parents



Inference in Bayesian networks Inference tasks

Simple queries: compute posterior marginal $\mathbf{P}(X_i|\mathbf{E}=\mathbf{e})$
e.g., $P(NoGas|Gauge=empty, Lights=on, Starts=$

Conjunctive queries: $\mathbf{P}(X_i, X_j|\mathbf{E}=\mathbf{e}) = \mathbf{P}(X_i|\mathbf{E}=\mathbf{e})\mathbf{P}(X_j$

Optimal decisions: decision networks include utility informati
probabilistic inference required for $P(outcome|actior$

Value of information: which evidence to seek next?

Sensitivity analysis: which probability values are most critical

Inference by enumeration

Slightly intelligent way to sum out variables from the joint without actually constructing its explicit representation

Simple query on the burglary network:

$$\mathbf{P}(B|j,m)$$
$$= \mathbf{P}(B,j,m)/P(j,m)$$
$$= \alpha\mathbf{P}(B,j,m)$$
$$= \alpha\ \Sigma_e\ \Sigma_a\ \mathbf{P}(B,e,a,j,m)$$

Rewrite full joint entries using product of CPT entries:

$$\mathbf{P}(B|j,m)$$
$$= \alpha\ \Sigma_e\ \Sigma_a\ \mathbf{P}(B)P(e)\mathbf{P}(a|B,e)P(j|a)P(m|a)$$
$$= \alpha\mathbf{P}(B)\ \Sigma_e\ P(e)\ \Sigma_a\ \mathbf{P}(a|B,e)P(j|a)P(m|a)$$

Consider the query $P(JohnCalls|Burglary = true)$

$$P(J|b) = \alpha P(b) \sum_e P(e) \sum_a P(a|b,e)P(J|a) \sum_m P(m|a)$$

Sum over m is identically 1; M is **irrelevant** to the query

Thm 1: Y is irrelevant unless $Y \in Ancestors(\{X\} \cup \mathbf{E})$

Here, $X = JohnCalls$, $\mathbf{E} = \{Burglary\}$, and $Ancestors(\{X\} \cup \mathbf{E}) = \{Alarm, Earthquake\}$
so $MaryCalls$ is irrelevant

(Compare this to backward chaining from the query in Horn cl

Defn: moral graph of Bayes net: marry all parents and drop arrows

Defn: $\mathbf{A}$ is m-separated from $\mathbf{B}$ by $\mathbf{C}$ iff separated by $\mathbf{C}$ in the mor

Thm 2: Y is irrelevant if m-separated from X by $\mathbf{E}$

7. Explain Dempster - Shafer Theory in AI (Apr-May'15)(Nov'15)

The **Dempster–Shafer theory (DST)** is a mathematical theory of evidence It allows one to combine evidence from different sources and arrive at a degree of belief (represented by a belief function) that takes into account all the available evidence. The theory was first developed by Arthur P. Dempster and Glenn Shafer.

In a narrow sense, the term **Dempster–Shafer theory** refers to the original conception of the theory by Dempster and Shafer. However, it is more common to use the term in the wider sense of the same general approach, as adapted to specific kinds of situations. In particular, many authors have proposed different rules for combining evidence, often with a view to handling conflicts in evidence better.

Dempster–Shafer theory is a generalization of the Bayesian theory of subjective probability; whereas the latter requires probabilities for each question of interest, belief functions base degrees of belief (or confidence, or trust) for one question on the probabilities for a related question.

These degrees of belief may or may not have the mathematical properties of probabilities; how much they differ depends on how closely the two questions are related Put another way, it is a way of representing epistemic plausibilities but it can yield answers that contradict those arrived at using probability theory.

Dempster–Shafer theory is based on two ideas: obtaining degrees of belief for one question from subjective probabilities for a related question, and Dempster's rule for combining such degrees of belief when they are based on independent items of evidence.

In essence, the degree of belief in a proposition depends primarily upon the number of answers (to the related questions) containing the proposition, and the subjective probability of each answer. Also contributing are the rules of combination that reflect general assumptions about the data.

In this formalism a **degree of belief** (also referred to as a **mass**) is represented as a **belief function** rather than a Bayesian probability distribution. Probability values are assigned to *sets* of possibilities rather than single events: their appeal rests on the fact they naturally encode evidence in favor of propositions.

Dempster–Shafer theory assigns its masses to all of the non-empty subsets of the entities that compose a system.

Belief and plausibility

Shafer's framework allows for belief about propositions to be represented as intervals, bounded by two values, *belief* (or *support*) and *plausibility*:

belief ≤ *plausibility*.

Belief in a hypothesis is constituted by the sum of the masses of all sets enclosed by it (*i.e.* the sum of the masses of all subsets of the hypothesis).[*clarification needed*]

It is the amount of belief that directly supports a given hypothesis at least in part, forming a lower bound. Belief (usually denoted *Bel*) measures the strength of the evidence in favor of a set of propositions. It ranges from 0 (indicating no evidence) to 1 (denoting certainty).

Plausibility is 1 minus the sum of the masses of all sets whose intersection with the hypothesis is empty. It is an upper bound on the possibility that the hypothesis could be true,

i.e. it "could possibly be the true state of the system" up to that value, because there is only so much evidence that contradicts that hypothesis.

Plausibility (denoted by *Pl*) is defined to be *Pl(s)=1-Bel(~s)*. It also ranges from 0 to 1 and measures the extent to which evidence in favor of *~s* leaves room for belief in *s*. For example, suppose we have a belief of 0.5 and a plausibility of 0.8 for a proposition, say "the cat in the box is dead." This means that we have evidence that allows us to state strongly that the proposition is true with a confidence of 0.5. However, the evidence contrary to that hypothesis (i.e. "the cat is alive") only has a confidence of 0.2.

The remaining mass of 0.3 (the gap between the 0.5 supporting evidence on the one hand, and the 0.2 contrary evidence on the other) is "indeterminate," meaning that the cat could either be dead or alive. This interval represents the level of uncertainty based on the evidence in your system.

Hypothesis	Mass	Belief	Plausibility
Null (neither alive nor dead)	0	0	0
Alive	0.2	0.2	0.5
Dead	0.5	0.5	0.8
Either (alive or dead)	0.3	1.0	1.0

The null hypothesis is set to zero by definition (it corresponds to “no solution”). The orthogonal hypotheses “Alive” and “Dead” have probabilities of 0.2 and 0.5, respectively. This could correspond to “Live/Dead Cat Detector” signals, which have respective reliabilities of 0.2 and 0.5. Finally, the all-encompassing “Either” hypothesis (which simply acknowledges there is a cat in the box) picks up the slack so that the sum of the masses is 1.

The belief for the “Alive” and “Dead” hypotheses matches their corresponding masses because they have no subsets; belief for “Either” consists of the sum of all three masses (Either, Alive, and Dead) because “Alive” and “Dead” are each subsets of

“Either”. The “Alive” plausibility is 1 − *m* (Dead) and the “Dead” plausibility is 1 − *m* (Alive). Finally, the “Either”

plausibility sums *m*(Alive) + *m*(Dead) + *m*(Either). The universal hypothesis (“Either”) will always have 100% belief and plausibility —it acts as a checksum of sorts.

Here is a somewhat more elaborate example where the behavior of belief and plausibility begins to emerge. We're looking through a variety of detector systems at a single faraway signal light, which can only be coloured in one of three colours (red, yellow, or green):

Combining beliefs

Beliefs corresponding to independent pieces of information are combined using *Dempster's rule of combination*, *which* is a generalization of the special case of Bayes' theorem where events are independent. Note that the probability masses from propositions that contradict each other can also be used to obtain a measure of how much conflict there is in a system. This measure has been used as a criterion for clustering multiple pieces of seemingly conflicting evidence around competing hypotheses.

In addition, one of the computational advantages of the Dempster–Shafer framework is that priors and conditionals need not be specified, unlike Bayesian methods, which often use a symmetry (minimax error) argument to assign prior probabilities to random variables (*e.g.* assigning 0.5 to binary values for which no information is available about which is more likely). However, any information contained in the missing priors and conditionals is not used in the Dempster–Shafer framework unless it can be obtained indirectly—and arguably is then available for calculation using Bayes equations.

8. Explain Fuzzy Logic in AI.(Apr-May'17)

Fuzzy logic is a form of many-valued logic or probabilistic logic; it deals with reasoning that is approximate rather than fixed and exact. In contrast with traditional logic theory, where binary sets have two-valued logic: true or false, fuzzy logic variables may have a truth value that ranges in degree between 0 and 1. Fuzzy logic has been extended to handle the concept of partial truth, where the truth value may range between completely true and completely false.[1] Furthermore, when linguistic variables are used, these degrees may be managed by specific functions.

Overview

The reasoning in fuzzy logic is similar to human reasoning. It allows for approximate values and inferences as well as incomplete or ambiguous data (fuzzy data) as opposed to only relying on crisp data (binary yes/no choices). Fuzzy logic is able to process incomplete data and provide approximate solutions to problems other methods find difficult to solve

Degrees of truth

Fuzzy logic and probabilistic logic are mathematically similar – both have truth values ranging between 0 and 1 – but conceptually distinct, due to different interpretations—see interpretations of probability theory. Fuzzy logic corresponds to "degrees of truth", while probabilistic logic corresponds to "probability, likelihood"; as these differ, fuzzy logic and probabilistic logic yield different models of the same real-world situations.Both degrees of truth and probabilities range between 0 and 1 and hence may seem similar at first. For example, let a 100 ml glass contain 30 ml of water. Then we may consider two concepts: Empty and Full. The meaning of each of them can be represented by a certain fuzzy set. Then one might define the glass as being 0.7 empty and 0.3 full.

Applying truth values

A basic application might characterize subranges of a continuous variable. For instance, a temperature measurement for anti-lock brakes might have several separate membership functions defining particular temperature ranges needed to control the brakes properly. Each function maps the same temperature value to a truth value in the 0 to 1 range. These truth values can then be used to determine how the brakes should be controlled.

Fuzzy logic temperature

In this image, the meanings of the expressions *cold*, *warm*, and *hot* are represented by functions mapping a temperature scale. A point on that scale has three "truth values"—one for each of the three functions. The vertical line in the image represents a particular temperature that the three arrows (truth values) gauge. Since the red arrow points to zero, this temperature may be interpreted as "not hot". The orange arrow (pointing at 0.2) may describe it as "slightly warm" and the blue arrow (pointing at 0.8) "fairly cold".

Linguistic variables

While variables in mathematics usually take numerical values, in fuzzy logic applications, the non-numeric *linguistic variables* are often used to facilitate the expression of rules and facts.

A linguistic variable such as *age* may have a value such as *young* or its antonym *old*. However, the great utility of linguistic variables is that they can be modified via linguistic hedges applied to primary terms. The linguistic hedges can be associated with certain functions.

The most important propositional fuzzy logics are:

- Monoidal t-norm-based propositional fuzzy logic MTL is an axiomatization of logic where conjunction is defined by a left continuous t-norm, and implication is defined as the residuum of the t-norm. Its models correspond to MTL-algebras that are prelinear commutative bounded integral residuated lattices.

– Basic propositional fuzzy logic BL is an extension of MTL logic where conjunction is defined by a continuous t-norm, and implication is also defined as the residuum of the t-norm. Its models correspond to BL-algebras.

– Łukasiewicz fuzzy logic is the extension of basic fuzzy logic BL where standard conjunction is the Łukasiewicz t-norm. It has the axioms of basic fuzzy logic plus an axiom of double negation, and its models correspond to MV-algebras.

– Gödel fuzzy logic is the extension of basic fuzzy logic BL where conjunction is Gödel t-norm. It has the axioms of BL plus an axiom of idempotence of conjunction, and its models are called G-algebras.

– Product fuzzy logic is the extension of basic fuzzy logic BL where conjunction is product t-norm. It has the axioms of BL plus another axiom for cancellativity of conjunction, and its models are called product algebras.

– Fuzzy logic with evaluated syntax (sometimes also called Pavelka's logic), denoted by

EVŁ, is a further generalization of mathematical fuzzy logic. While the above kinds of fuzzy logic have traditional syntax and many-valued semantics, in EVŁ is evaluated also syntax. This means that each formula has an evaluation. Axiomatization of EVŁ stems from Łukasziewicz fuzzy logic. A generalization of classical Gödel completeness theorem is provable in EVŁ.

9. Explain Certainty factors and Rule based systems .(Apr-May '16)

Certainty factors:

- The certainty-factor model was one of the most popular model for the representation and manipulation of uncertain knowledge in the early (1980s) Rule-based expert systems.

- The model was criticized by resea hers in artificial intelligence and statistics being ad- hoc-in nature. Resea hers and developers have stopped using the model.
- Its place has been taken by more expressive formalisms of Bayesian belief networks for the representation and manipulation of uncertain knowledge.
- The manipulation of uncertain knowledge in the Rule-based expert systems is illustrated in the next three slide before moving to Bayesian Networks.

Rule Based Systems:

Rule based systems have been discussed in previous lectures.

Here it is recalled to explain uncertainty.A rule is an expression of the form "if A then B" where A is an assertion and B can be either an action or another assertion.

Example : Trouble shooting of water pumps

- If pump failure then the pressure is low
- If pump failure then check oil level
- If power failure then pump failure

Rule based system consists of a library of such rules.Rules reflect essential relationships within the domain.

Rules reflect ways to reason about the domain.Rules draw conclusions and points to actions, when specific information about the domain comes in. This is called inference.

The inference is a kind of chain reaction like :

If there is a power failure then (see rules 1, 2, 3 mentioned above) Rule 3 states that there is a pump failure, and

Rule 1 tells that the pressure is low, and

Rule 2 gives a (useless) recommendation to check the oil level.

■ It is very difficult to control such a mixture of inference back and forth in the same session and resolve such uncertainties.

How to deal such uncertainties ?

How to deal uncertainties in rule based system?

A problem with rule-based systems is that often the connections reflected by the rules are not absolutely certain (i.e. deterministic), and the gathered information is often subject to uncertainty.

In such cases, a certainty measure is added to the premises as well as the conclusions in the rules of the system.

A rule then provides a function that describes : how much a change in the certainty of the premise will change the certainty of the conclusion.

In its simplest form, this looks like :

If A (with certainty x) then B (with certainty f(x))

This is a new rule, say rule 4, added to earlier three rules.

There are many schemes for treating uncertainty in rule based systems. The most common are :

- Adding certainty factors.
- Adoptions of Dempster-Shafer belief functions.
- Inclusion of fuzzy logic.

In these schemes, uncertainty is treated locally, means action is connected directly to incoming rules and uncertainty of their elements. Example : In addition to rule 4 , in previous slide, we have the rule If C (with certainty x) then B (with certainty g(x))

QUESTION BANK & UNIVERSITY QUESTIONS

1. Consider the following price of knowledge.

A1: Tony, Mike and John belong to the Alpine Club.

A2: Every member of the Alpine Club who is not a skier is a mountain climber.

A3: Mountain Climber does not like rain and anyone who does not like show is not a skier.

A4: Mike dislikes whatever Tony likes and like whatever Tony dislikes.

A5: Tony likes rain and show.

Can you conclude “There is a member of the Alpine Club who is a mountain climber but not a skier”? Justify your answer.

2. How is knowledge represented and reasoning done for “SCRIPTS”?
3. Write short notes on:
    a. Truth Maintenance.
    b. Probabilistic Reasoning
4. Explain the various steps associated with the knowledge engineering process. Discuss by applying the steps to any real world problem
5. Explain probabilistic reasoning systems in detail. (Apr-May’15)(Dec 17)((Pg. No. 17, Q. No. 3)
6. (a) Explain sequential decision problems in detail.

    (b) Write in detail about Truth Maintenance.]
7. Discuss the semantics Bayesian network in detail with an example. (Nov’13)(Nov’15)(April’18) (Pg. No. 23, Q. No. 6)
8. Discuss the following in detail:
    a. Multi attribute utility functions.
    b. Nonmonotonic reasoning. (Apr-May’14)(Nov-Dec’14)(Nov’15)(Pg. No 15, Q. No. 2)
9. Describe the Dempster – Shaffer Theory in Detail (Apr- May’15)(Nov’15)(Dec ’17)(Nov’18)(Pg. No. 31, Q. No. 7
10. Explain Certainty factors and Rule based systems .(Apr-May ’16)
11. Explain Fuzzy Logic in AI.(Apr-May’17) (Apr –May ’16)(April 18)
12. What is the need for probability theory in uncertainty? Consider the following fictitious scientific information. Doctors find that the people with disease KJ almost invariably ate hamburgers, thus P(HamburgerEater) = 0.6, What is the probability that a hamburger eater will have KJ disease?(Nov’18)